

ABI Hotel Booking System

Project Documentation

Project Overview

ABI Hotel Booking System is a full-stack MERN application that enables customers to browse hotels, search, view rooms, make bookings, complete payments, and review booking history. An admin dashboard provides operational insights and management screens.

Architecture

Frontend: React, React Router, Axios, Bootstrap, Material UI

Backend: Node.js, Express.js

Database: MongoDB Atlas with Mongoose

Deployment: Render (Frontend & Backend), MongoDB Atlas

Key Features

- Hotel listing and search
 - Hotel details with available rooms
 - Room booking workflow
 - Guest capacity validation
 - Date-overlap booking validation
 - Payment simulation and confirmation
 - My Bookings page
 - Admin Dashboard
 - Manage Rooms
 - Booking Management
 - Customer Management
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Business Logic

Date Overlap Validation

The system prevents double booking by checking whether the requested booking dates overlap with existing confirmed bookings.

Guest Capacity Validation

Bookings are rejected when the requested guest count exceeds the selected room's capacity.

Application Flow

Home → Hotel Details → Room Selection → Booking Form → Payment → Booking Confirmation → My Bookings

Admin Module

Dashboard displaying total hotels, rooms, bookings, customers, and revenue. Separate management pages are available for rooms, bookings, and customers.

Deployment

Frontend: <https://abi-hotel-booking-1.onrender.com>

Backend API: <https://abi-hotel-booking.onrender.com>

Repository: <https://github.com/Rahul-PuriRepo/abi-hotel-booking>

Assumptions

- Payment is simulated for demonstration purposes.
 - MongoDB Atlas stores hotel, room, customer, and booking data.
 - Revenue is calculated from successful bookings.
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Future Improvements

- JWT Authentication
- Role-based access control
- Online payment gateway integration
- Hotel management module
- Email notifications
- Reviews and ratings

Note: Screenshots can be added as an appendix if required for submission.